

FlyRank AI Internship

Machine Learning Track

WEEK 01 REPORT

Running the Starter Notebooks & Initial Data Discovery

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Track	Machine Learning
Week	01
Repository	GitHub Repository
Status	Completed

1. 1. Executive Summary

During the first week of the FlyRank Machine Learning Internship, I successfully completed the **"Run the Starter Notebooks (ML-01)"** assignment using **Google Colab** and **GitHub**. The objective of this assignment was to gain practical exposure to a complete machine learning workflow rather than focusing only on theoretical concepts.

Throughout the notebook, I executed an end-to-end machine learning pipeline on an anonymized real-world SEO dataset containing 30,000 records. The workflow included data preparation, feature engineering, baseline model evaluation, training multiple machine learning models, comparing their performance, and generating reports automatically.

In addition to running the complete pipeline, I explored the dataset using Pandas, analyzed important performance metrics, and performed an independent data discovery exercise to better understand the relationship between search volume and page impressions. This assignment provided valuable hands-on experience with machine learning pipelines, data analysis, and model evaluation while establishing a strong foundation for the remaining internship activities.

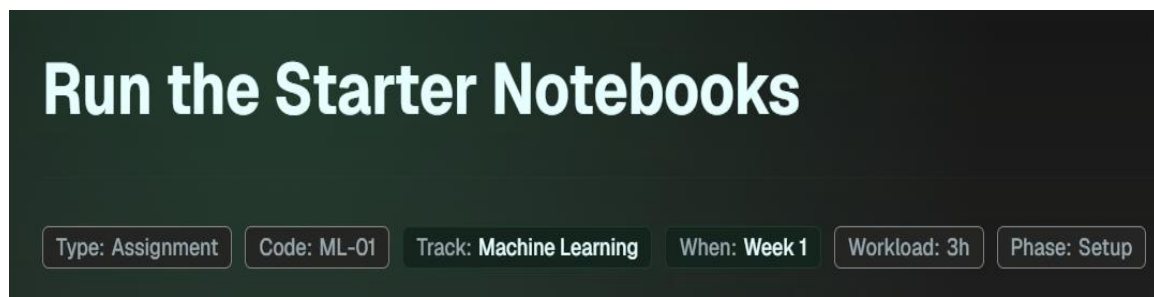


Figure 1: Assignment Overview from the FlyRank InternHQ Portal.

2. Assignment Objectives

The primary objectives of this assignment were to:

- Execute the provided starter notebooks successfully in Google Colab.
- Understand the complete workflow of a machine learning pipeline.
- Explore and analyze an anonymized real-world SEO dataset.
- Compare the performance of a rule-based baseline with trained machine learning models.
- Perform an independent exploratory data analysis to derive meaningful insights.
- Save the executed notebooks to a personal GitHub repository for version control and future submissions.

3. Development Environment & Tools

The following technologies and tools were used throughout this assignment:

Tool	Purpose
Google Colab	Executing Python notebooks in the cloud
GitHub	Repository management and assignment submission
Python	Primary programming language
Pandas	Data loading, cleaning, and analysis
NumPy	Numerical computations
Scikit-learn	Machine learning model training and evaluation

The notebook was executed entirely in Google Colab without requiring any local software installation.

4. Machine Learning Pipeline Overview

The provided notebook automatically executed a complete machine learning workflow consisting of the following stages:

Step 1 – Data Preparation & Feature Engineering

The raw SEO dataset was processed into a structured feature vector suitable for machine learning. Relevant features were prepared while defining the target variable for prediction.

Step 2 – Baseline Rule

A transparent rule-based model was created to establish a benchmark. This baseline represents a manually designed decision rule against which machine learning models could be compared.

Step 3 – Model Training

Three different supervised learning algorithms were trained using the prepared dataset:

- Logistic Regression
- Decision Tree
- Random Forest

Each model learned patterns from the training data to predict which webpages were likely to decline in search performance.

Step 4 – Model Evaluation

After training, each model was evaluated using performance metrics such as **Precision@50**. The models were compared to determine which produced the most accurate predictions.

Step 5 – Report Generation

Finally, the notebook automatically generated:

- Prediction outputs
- Performance reports
- Evaluation charts
- A PDF summary of the complete machine learning pipeline

```
=====
▶ Step 1/5 – Prepare features – clean the data, build the feature vector, define the label
=====
Prepared 30,000 rows from 30,000 raw rows
Wrote /content/flyrank-ml-internship-starter/data/processed/refresh_feature_vector.csv

=====
▶ Step 2/5 – Baseline – a transparent hand-written rule to beat
=====
Wrote baseline queue: /content/flyrank-ml-internship-starter/data/processed/baseline_refresh_queue.csv
Top-50 declining rate (full data, not the evaluated holdout Precision@50): 0.340

=====
▶ Step 3/5 – Train – logistic regression, decision tree, random forest (client-holdout split)
=====
Trained 3 models on 30,000 rows
Split strategy: client_holdout
Best model: random_forest
Wrote predictions: /content/flyrank-ml-internship-starter/data/processed/model_predictions.csv
Wrote model results: /content/flyrank-ml-internship-starter/outputs/model_results.json

=====
▶ Step 4/5 – Evaluate – ranked refresh queue, charts, and the Markdown report
=====
Wrote final refresh queue: /content/flyrank-ml-internship-starter/outputs/refresh_queue.csv
Wrote model report: /content/flyrank-ml-internship-starter/outputs/model_report.md
Wrote charts in: /content/flyrank-ml-internship-starter/outputs/charts

=====
▶ Step 5/5 – Report – a shareable PDF summary
=====
Wrote PDF report: /content/flyrank-ml-internship-starter/outputs/flyrank_refresh_model_results.pdf

Pipeline complete
```

Figure 2: Successful execution of the complete ML pipeline showing all five processing stages.

5. Model Performance Results

The notebook compared the performance of a manually designed baseline rule with a trained Random Forest model.

Model	Precision@50
Baseline Rule	0.240 (~12 correct predictions out of 50)
Random Forest	0.740 (~37 correct predictions out of 50)

The Random Forest model achieved approximately **3.1 times higher Precision@50** than the baseline rule. This demonstrates how machine learning models can identify complex patterns in data that are difficult to capture using manually written rules.

These results highlight the effectiveness of supervised learning techniques when applied to real-world datasets.

```
Hand-written rule Precision@50: 0.240 (~12 of the top 50 right)
Random forest Precision@50: 0.740 (~37 of the top 50 right)

The learned model roughly 3.1x the rule on this metric.
Validation split used: client_holdout (pages from a client are never in both train and test)
```

Figure 3: Performance comparison between the Baseline Rule and Random Forest model.

6. Exploratory Data Analysis (EDA)

Several exploratory analyses were performed to better understand the characteristics of the dataset.

Observation 1 – CTR by Search Position

The average Click-Through Rate (CTR) was analyzed across different search result positions.

The analysis revealed a clear downward trend:

- Pages ranking on the first page received the highest CTR.
- CTR gradually decreased as ranking positions became lower.
- Deep search result pages received the lowest user engagement.

This observation reinforces the importance of achieving higher search rankings for improving website traffic.

Observation 2 – Content Length

The median word count of growing and declining pages was compared.

The results showed only a small difference between the two groups, suggesting that content length alone is not a reliable indicator of search performance.

```
print(df.shape[0], 'rows', df.shape[1], 'columns')
df.head(3)
```

30000 rows, 44 columns																					
	content_id	client_id	search_volume	competition	competition_level	cpc	content_type	main_intent	word_count	char_count	...	char_count_tier	ctr	avg_position	engagement_rate	scroll_rate	ai_traffic_pct	impression_tier	position_tier	trend_direction	trend_pct
0	content_30440230142	client_06924896	10.0	0.67	HIGH	2.05	keyword article	transactional	3221.0	20457.0	...	15000-25000	0.76	10.6	5.88	4.55	0.0	good	striking	down	-41.4
1	content_11b4e703a9e	client_4e07408562	90.0	0.01	LOW	0.05	keyword article	informational	2481.0	15562.0	...	15000-25000	0.05	20.3	0.00	10.00	0.0	good	page_3_5	down	-57.7
2	content_8aa7934a0895	client_7225307e2	0.0	0.00	LOW	0.00	keyword article	informational	3515.0	23643.0	...	15000-25000	0.09	36.5	0.00	28.57	0.0	good	page_3_5	down	-60.9

3 rows x 44 columns

Figure 4: Dataset preview generated using `df.head()`.

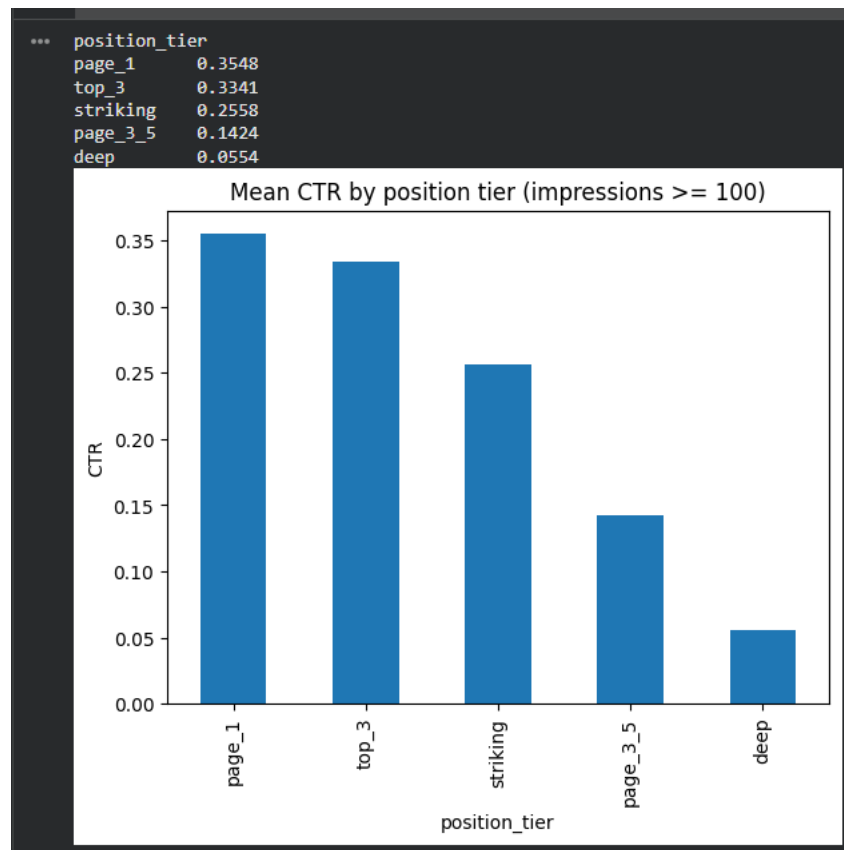


Figure 5: Mean CTR across different search ranking positions.

7. Independent Data Discovery

As part of the assignment, I selected the first optional discovery task.

The original notebook examined the relationship between **search volume** and **page impressions**. I repeated this analysis after filtering the dataset to include only pages with **impressions greater than zero**.

Result

New Correlation = 0.001

Interpretation

The calculated correlation remained extremely close to zero even after filtering the dataset.

This suggests that keyword search volume alone has almost no linear relationship with the actual number of page impressions within this dataset.

Instead, page visibility appears to be influenced by multiple factors such as:

- Search ranking position
- Click-through rate (CTR)
- Content quality
- User engagement
- Overall SEO performance

This exercise demonstrated the importance of validating assumptions using data rather than relying solely on common SEO beliefs.

```
Correlation between search_volume and impressions_90d: 0.001  
Near zero -> keyword search volume barely predicts the traffic a page actually gets.
```

```
New Correlation: 0.0012030310696208905
```

Figure 6: Custom discovery code and output showing the recalculated correlation.

9. Skills Acquired

By completing this assignment, I strengthened both my technical knowledge and practical machine learning skills.

Technical Skills

- Python Programming
- Data Analysis using Pandas
- NumPy
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Machine Learning Fundamentals
- Random Forest Classification
- Model Evaluation
- Google Colab
- GitHub Version Control

Professional Skills

- Analytical Thinking
 - Problem Solving
 - Data Interpretation
 - Technical Documentation
 - Research-Based Learning
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10. Reflection

This assignment provided an excellent introduction to practical machine learning by allowing me to work with a real anonymized dataset and execute a complete end-to-end ML pipeline. Rather than simply learning theoretical concepts, I experienced how data is prepared, models are trained, evaluated, and compared using objective performance metrics.

The independent discovery task encouraged critical thinking and reinforced the importance of validating assumptions through data analysis. Running the notebook, interpreting the outputs, and documenting the findings significantly improved my understanding of machine learning workflows and increased my confidence in using Python, Google Colab, and GitHub for AI projects.

Overall, Week 1 established a solid foundation for the remainder of the FlyRank Machine Learning Internship and prepared me to tackle more advanced machine learning concepts and real-world applications in the upcoming weeks.

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